BM
3/22/22

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES

DATE COLLECTED: 14 MAR 2022 Initials. TW

Conversion Factors
10 in. Bulk Rain Funnel: NET divided by 51.1 = catch
N/A if Overfilled
12 in Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	CATCH (mm)	ETI CATCH (mm)	Coll. EFF. (% of ETI)
RG-22	1200	701	10.1		
RG-1	1015	740	10.7		
RG-4	1035	585	8.4		
RG-23	0855	834	12.0		

Volumes Collected

		RG-22	RG-1	RG-4	RG-23
Gross	g	1661	1700	1547	1795
Tare	g	960	960	962	961
Net	g	701	740	585	834

SITE CONTAMINATION

(0 = none.1 = low ------------------------> 5 = high)

SITE	Fine PM	Coarse PM	Bird FECES	Insects	Color	GB Ring	dBASE FIELD code?	SAMPLE REJECTED @LEVEL =	Database CODES checked ?
RG-22	4	3 seeds Bark	1 Bucket hit	0	0	3	969		
RG-1	3	1 "	0	0	0	2			
RG-4	2	1 pine needle	0	0	0	1			
RG-23	3	0	0	0	0	2			

Precipitation amount
.81 inches / 20.6 mm
ETI collector

ADDITIONAL COMMENTS: Precip consisted of rain and snow, refrozen @ collection. STA/22 SG= 13", SG-1= 4" (Disturbed) SG-4= 15", SG-23= 22"
FS snow data offline

Applicable dBASE FIELD codes: (See SOP Manual)
970 = Obvious Contamination: Sample REJECTED
969 = Possible Contamination. Check Return Analyses
955 = QNS: Quantity Not Sufficient for analyses (< 100 mL NET catch)

HUBBARD BROOK ECOSYSTEM STUDY **SAMPLE NOTES**

Streamwater Samples

Date Collected: 14 MAR 2022 Initials. TW

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none:1 = low---->5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	1010	0.1°	.252	L	0	0	Mostly se/ic	
WS-2	1020	0.2°	.234	L	0	0	"	
WS-3	1045	0.6°	.341	L	0	0	"	
WS-4	1000	0.1°	.320	L	0	0	"	
WS-5 (Dup)	0940	0.1°	.342	L	0	0	<50% opw w/ ice/snow.	
WS-6	0950	0.2°	.254	L	1-	0	Mostly ic/sc	
WS-7	0920	0.2°	.377	L	0	0	one opw hole.	
WS-8	0905	0.3°	.315	L	1	0	"	
WS-9	0850	0.1°	.293	L	2-	0	No opw.	
Hubbard BK	1110	0.1°		L	1-	0	Collapsed ice shelves, opw pools, OTTER TRACKS	
Mlake Outlet	1140	2.8°		F	0	0	opw staff, OTB	

ADDITIONAL COMMENTS: Fresh few inches of powder. Streams have some opw, w/ ice/snow in channels. Bank ice in-tact, Some weirs frozen, some open.

Applicable dBASE FIELD codes: 911= High Flow Event in Progress; 912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)
955 = QNS: Quantity Not Sufficient for analyses

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 15 MAR 2022 - 3* Initial pH
25 MAR 2022 - 3* re-run + Metrohm RUN By: TW

Meter CALIBRATION: See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.14x at: 17.8 °C Lab Temp

Precipitation

Samples:	DATE COLLECTED	Metrohm pH		3 Star pH	SpCond @ 25°C	Remarks Codes
STATION	14 MAR 2022	NA	Run 3/14/22	5.66	7.2	
RG-1		No Data		5.43	4.9	
RG-4				5.38	5.5	
RG-23	14 MAR 2022	NA		5.39	4.4	

Streamwater

Samples	DATE COLLECTED	Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH (3/14)	3 Star pH 3-25	SpCond @ 25°C	Remarks Codes
WS-1	14 MAR 2022	5.58	+6.9	5.36	(5.55)	5.53	13.9	
WS-2		5.53	+5.8	5.24	(5.57)	5.57	8.8	
WS-3		5.79	+13.9	5.07	(5.80)	5.83	10.6	
WS-4		6.02	+17.8	5.13	(6.05)	6.07	10.2	
WS-5		5.47	+4.3	5.20	(5.53)	5.54	11.8	
WS-6		5.63	+6.4	5.38	(5.64)	5.64	11.7	
WS-7		6.11	+21.4	5.39	(6.13)	6.12	11.0	
WS-8		5.93	+16.4	5.21	(5.95)	5.98	11.1	
WS-9		4.91	−23.6	4.51	(4.97)	4.92	14.0	
Hubbard BK		6.31	+34.0	5.02	(6.33)	6.32	11.7	
Mlake Outlet	14 MAR 2022	6.32	+67.8	5.18	(6.21)	6.29	35.5	

ADDITIONAL COMMENTS: Metrohm computer down. Samples refrigerated for later run. 3* pH initial run in parentheses. Resumed titrations for this sample batch 3/25/22. Could not run Metrohm pH on precip. Buckets were dumped, washed for following week's catch.

Applicable codes: 911= High Flow Event in Progress;
955= QNS. Quanity Not Sufficient for analyses(<100 mL NET catch)
912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

DIC-[date] xls <-save as [date]= mm-dd-yy

HBEF Stream and M.Lake Routine Sample DIC Data

date: **14MAR2022** Samples collected 03/14/2022

Aqueous STANDARDS

type	DIC set =	peak ht mm	adjusted mm	ATT -X
DIW blank	0	10	0	16
	0	10	0	16
	0	12	0	16
100 µM	100	32	22	16
	100	30	20	16
	100	30	18	16
200 µM	200	51	41	16
	200	51	41	16
	200	52	40	16
400 µM	400	95	85	16
	400	95	85	16
	400	93	81	16

Statistics *Values from matrix at M5 -->*

slope =	4 761
slope ±	0.067
Y-int =	3 2
Y-int ±	3 2
R-sq =	0 9980

Laboratory Conditions

Lab Temp =	14.6	°C
Barom. Pressure =	774	mm Hg
Analyst's Initials	TW	50%RH
NaCO3 Standards		
0 1M Stock Dated ·		3/10/22
0 01M Working Solution Dated.		3/10/22

Samples	inj 1 1	inj 2 2	inj 3 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC µM/L	Calc err±	NOTES 120 mA current
ML-70	43	43	42	16	42 7	0.6	ML-70	**203**	3	
	0	0	0	16	0.0	0.0	0	**0**	0	
	0	0	0	16	0 0	0 0	0	**0**	0	
	0	0	0	16	0 0	0 0	0	**0**	0	
	0	0	0	16	0 0	0.0	0	**0**	0	
W-1	15	16	21	16	17.3	3.2	W-1	**83**	15	
W-2	12	13	13	16	12 7	0.6	W-2	**60**	3	
W-3	12	11	12	16	11.7	0.6	W-3	**56**	3	
W-4	12	12	13	16	12.3	0 6	W-4	**59**	3	
W-5	10	10	10	16	10 0	0.0	W-5	**48**	0	
W-6	11	10	10	16	10.3	0.6	W-6	**49**	3	
W-7	14	14	13	16	13 7	0.6	W-7	**65**	3	
W-8	13	13	12	16	12.7	0 6	W-8	**60**	3	
W-9	10	10	12	16	10 7	1.2	W-9	**51**	5	
HBk	15	15	15	16	15.0	0 0	HBk	**71**	0	
	0	0	0	16	0 0	8 0	0			
							0			
Lab Air	15	14	15	16	14.7	0.6	Lab Air	**70**	3	
Outside Air	12	11	11	16	11 3	0 6	Outside Air	**54**	3	
Gas 1000	25	25	27	16	25.7	1 2	Gas 1000	**122**	5	std=83.3
Gas 3000	75	75	76	16	75 3	0 6	Gas 3000	**359**	3	*std=246 3*